Lightweight RF Fingerprint Identification for LEO Satellites Using Short-Duration Signals

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***Abstract*—Radio Frequency Fingerprint (RFF) identification is critical for space target monitoring and anti-spoofing. However, the high mobility, significant Doppler shifts, and non-stationary nature of Low Earth Orbit (LEO) satellite signals make robust fingerprint extraction highly challenging in non-cooperative environments. this paper proposes a lightweight RFF identification approach pioneering the use of short-duration Iridium Unique Word (UW) signals. Featuring a deterministic position, fixed duration, and constant modulation, the UW effectively captures inherent hardware fingerprints. By integrating a unified preprocessing workflow with a generalized deep learning model, this method significantly reduces signal processing complexity and computational overhead. Experimental results using real-world data from 66 Iridium satellites demonstrate that the 0.48-ms UW segment effectively carries individual identity features, achieving an accuracy of 83.6%. This approach provides a low-complexity solution for LEO satellite identification in data-constrained scenarios.**

Keywords—Specific Emitter Identification, Low Earth Orbit Satellites, Short-Time Signals, Iridium Unique Word, Deep Learning.

I. Introduction

The rapid expansion of Low Earth Orbit (LEO) satellite constellations is fundamentally reshaping global communications. Owing to their low orbital altitude and wide coverage, LEO satellites play pivotal roles in remote sensing, navigation, and emergency services [1]. However, their highly dynamic motion complicates continuous individual tracking, underscoring the urgent need for robust Radio Frequency Fingerprint (RFF) identification techniques.

Existing RFF methods generally fall into traditional feature engineering and deep learning categories. Traditional approaches rely on hand-crafted temporal, spectral, or statistical features [2]. Recently, deep learning has gained prominence for its capacity to automatically extract subtle fingerprints. For instance, PAST-AI converts signals into grayscale images for 66-satellite classification [3], while 3D-CNNs model temporal characteristics [4]. Further advances include feature enhancement [5], deep clustering [6], and cross-modal modeling [7].

However, most methods assume structurally complete signals. In realistic non-cooperative scenarios, captured waveforms are frequently fragmented by limited observation windows and corrupted by noise and fading [8], severely impairing feature stability. Moreover, achieving high accuracy often demands complex architectures, hindering deployment in resource-constrained settings. Hence, a low-complexity RFF identification approach is critically needed.

To address these challenges, this paper proposes a lightweight RFF identification framework targeting Iridium satellites. The main contributions are summarized as follows:

- The short-duration Iridium Unique Word (UW) segment is adopted for RFF extraction. Its deterministic temporal position and constant BPSK modulation eliminate the reliance on complex protocol parsing and complete frame structures. Additionally, the fixed baseband symbols, independent of stochastic user payloads, provide a highly stable carrier for hardware-specific impairments.
- A tailored preprocessing pipeline is designed to mitigate the significant Doppler shifts and noise interference inherent to received signals in non-cooperative scenarios.
- 3A dedicated ResNet-34-1D identification model is developed specifically for short-duration I-Q sequences. Validation on a real-world dataset comprising 66 Iridium satellites demonstrates that the proposed method achieves effective identification utilizing merely 0.48-ms signal segments.

The remainder of this paper is organized as follows. Section II details the preprocessing pipeline. Section III introduces the classification model. Section IV presents the experimental results and analysis, and Section V concludes the paper.

II. Iridium Signal Preprocessing Methodology

This section describes the Iridium downlink signal structure. It also details the targeted preprocessing pipeline for constructing the UW dataset used by the identification model. The framework of the individual identification system based is shown in Fig. 1.

A. Iridium Signal Structure

The Iridium radio signals are transmitted in the L-band, in the frequency range 1616.0–1626.5 MHz, utilizing Time Division Multiple Access to divide 90-ms frames into distinct time slots. Each simplex channel slot transmits a burst signal lasting 6.5–20.32 ms comprising a tone pilot, a UW, and a data payload. The burst signal generation is shown in Fig. 2. The UW, primarily used for phase correction [9], is modulated

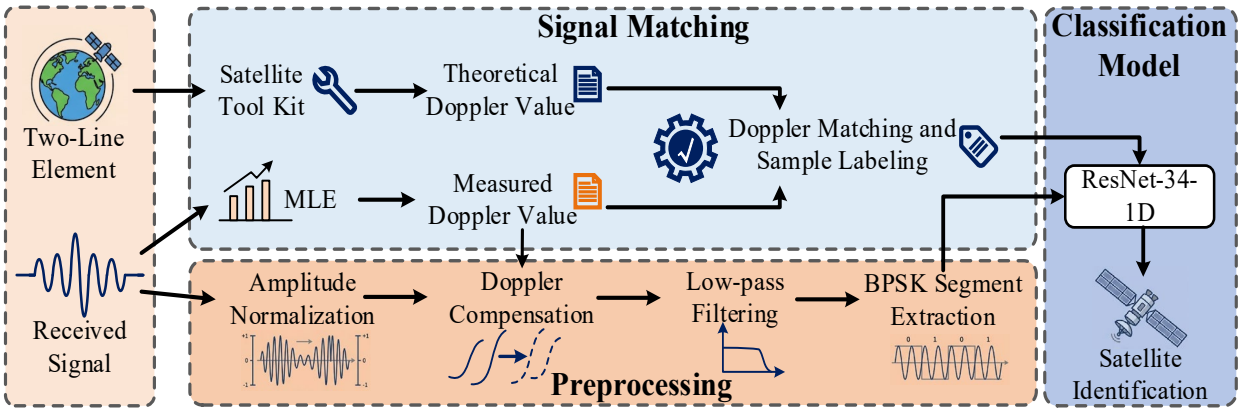

Fig. 1. Overall framework of the proposed method.

with a fixed 12-bit baseband sequence ("789" in hexadecimal). Because this sequence is independent of user payloads, it provides an exceptionally stable carrier for hardware-specific impairments. Consequently, we select the UW from the simplex channel as the RFF carrier.

B. Signal acquisition and data preprocessing

In non-cooperative scenarios, received signals suffer from unknown onsets, large frequency offsets, and noise [10]. To ensure high-quality inputs and accurate labels for the neural network, a systematic preprocessing framework is implemented.

- Signal Acquisition: Iridium receiving platform illustrated in Fig. 3. Downlink signals are captured via an L-band antenna, amplified by a Low Noise Amplifier (LNA), and down-converted by a receiver [11]. Digitized I-Q samples are stored at a 960 kHz sampling rate.
- Burst Detection: Due to the absence of prior timing information, matched filtering is applied to the unmodulated, highly concentrated preamble tone at the frame's start to detect the signal onset and align the frames [12].
- Doppler Estimation: LEO satellite mobility induces severe Doppler shifts [13], degrading modulation features [14]. To compensate, a Fast Fourier Transform extracts a coarse frequency, followed by a Maximum Likelihood Estimation for fine-grained accuracy. The likelihood function is defined as:

$$M(f) = \frac{1}{N} \cdot \left| \sum_{n=0}^{N-1} x[n] \cdot \exp(-j \cdot 2\pi \times f \cdot n) \right|^2 \quad (1)$$

  where f represents the frequency to be estimated, N is the number of sampling points, and $x(n)$ denotes the n signal sample.
- Normalization and Low-pass Filtering: To suppress amplitude variations and out-of-band interference, the aligned signals are first normalized and subsequently pass through passed through a low-pass filter .
- Signal Labeling: Operating non-cooperatively, we adopt a Doppler matching approach. The measured Doppler estimates are matched against theoretical frequency trajectories generated via Satellite Tool Kit using reception timestamps, thereby accurately assigning satellite identity labels [15].

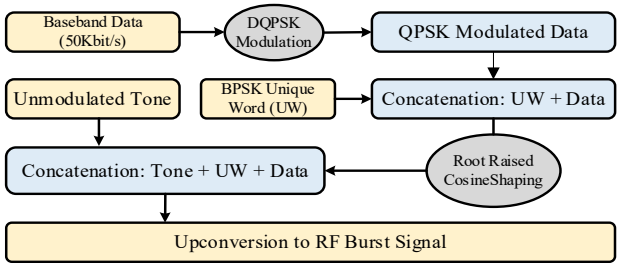

Fig. 2. Generation process of the Iridium downlink burst signal..

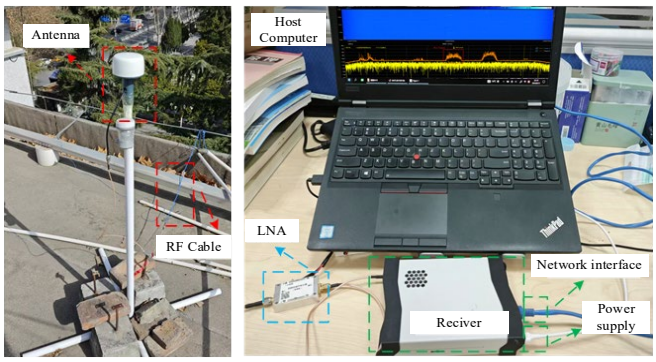

Fig. 3. The experimental Iridium signal receiving platform.

III. Cassification Model: ResNet-34-1D

To characterize subtle individual differences in short-duration 1D I-Q UW signals, we propose a tailored ResNet-34-1D model based on the ResNet-34 architecture [16]. To accommodate time-series data, all standard 2D convolution and pooling operations are replaced with 1D counterparts. Furthermore, to rapidly capture highly concentrated discriminative features, we implement an asymmetric kernel optimization strategy within the residual blocks. Specifically, the first convolutional layer retains a kernel size of 3 to extract local transient features, while the second layer's kernel is expanded to 21. This significantly broadens the receptive field for modeling extended time-scale information. By leveraging shortcut connections to prevent vanishing gradients, this architecture perfectly balances local sensitivity with long-range dependencies, ensuring the robust and computationally efficient extraction of instantaneous hardware characteristics. ResNet-34-1D model is illustrated in Fig. 4.

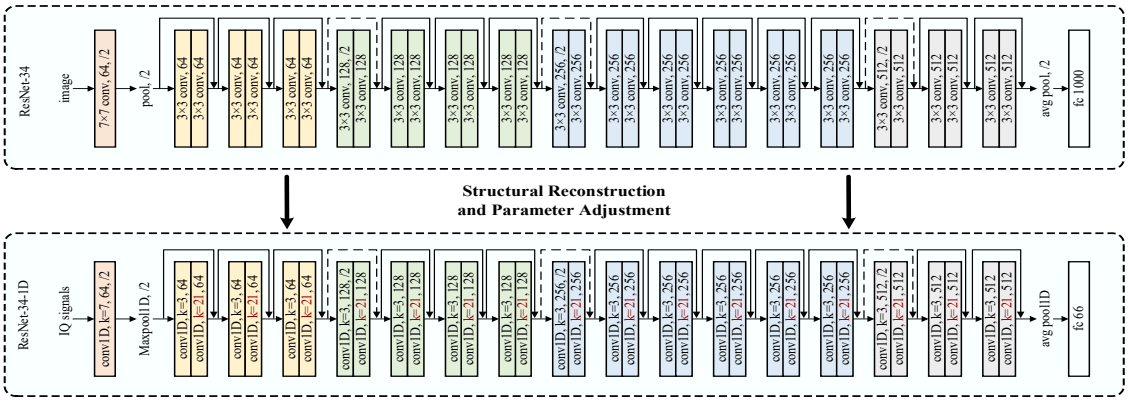

Fig. 4. ResNet-34-1D architecture.

TABLE I. PERFORMANCE COMPARISON OF INDIVIDUAL IDENTIFICATION METHODS

Method	*Accuracy*	*Sample size*	*Sample length (ms)*	*Modulation Scheme*	*Demodulation Required*	*Data Format*
PAST-AI[3]	83.0%	198,000,000	3.42–12.92	QPSK	Yes	Image
3D-CNN[4]	92.3%	198,000,000	3.42–12.92	QPSK	Yes	Image
ResNet-34-1D (Ours)	71.8%	261,070	3.125	QPSK	No	I-Q Sequence
ResNet-34-1D (Ours)	**83.6%**	**261,070**	**0.48**	**BPSK**	**No**	**I-Q Sequence**

IV. EXPERIMENTS

A. Settings

Iridium signals were continuously collected for four days using the receiving platform illustrated in Fig. 3, yielding a dataset comprising 283,595 I-Q samples from 66 Iridium satellites. This dataset was partitioned into training, validation, and testing sets. Fig. 5 and Fig. 6 illustrate the sample quantity and signal-to-noise ratio (SNR) distributions across the 66 satellite categories, respectively. Following the aforementioned preprocessing pipeline, a 460-point BPSK segment and a 3,000-point QPSK segment were extracted from each sample to conduct comparative experiments evaluating different modulation segments.

The ResNet-34-1D network was implemented in PyTorch. Training utilized the cross-entropy loss function and the Adam optimizer with an initial learning rate of 0.0001. The batch size was set to 64, and the model was trained for 80 epochs.

B. Performance Evaluation and Discussions

Fig. 7 and Fig. 8 present the confusion matrices for the 66-class classification task, respectively. Under identical conditions, the average recognition accuracy of the BPSK segments reaches 83.6% with clear classification boundaries, outperforming the 71.8% achieved by the longer-duration QPSK segments. This performance gap confirms that the invariant BPSK-modulated UW avoids interference from stochastic user data, making it a reliable carrier for hardware-specific impairments. Furthermore, the t-SNE visualization of the UW, Fig. 9 exhibits distinct intra-class compactness and inter-class separation., This clustering morphology validates that the BPSK UW intrinsically provides highly discriminative RFF features even under channel noise.

TABLE I. compares the proposed framework with classical methods (PAST-AI and 3D-CNN). The proposed method utilizes approximately 230,000 samples—orders of magnitude fewer than the 198-million-scale datasets required by the reference methods. It relies on UW signals lasting merely 0.48-ms, whereas the comparative methods use QPSK segments ranging from 3.42 to 12.92 ms. Additionally, these counterparts demand high SNR conditions (e.g. 9 dB for PAST-AI [3]), rigorous signal demodulation, and image conversion. In contrast, the proposed method bypasses complex preprocessing to directly process the raw I-Q sequences. Although the classical methods achieve high accuracy, their stringent prerequisites limit their practical feasibility. Conversely, the proposed method successfully identifies 66 individual satellites under data-constrained, non-cooperative conditions, fully validating its substantial practical value.

V. CONCLUSION

This paper proposes a lightweight identification strategy leveraging short-duration Iridium UW signals. Integrating a tailored preprocessing pipeline with a generalized network achieves effective individual identification, outperforming longer QPSK segments under identical conditions. While first to demonstrate UW's feasibility in Iridium RFF identification, its absolute accuracy has yet to surpass existing methods. Future research will prioritize network optimization and feature fusion for RFF extraction in complex cross-modulation environments, extending this methodology to large-scale LEO constellations.

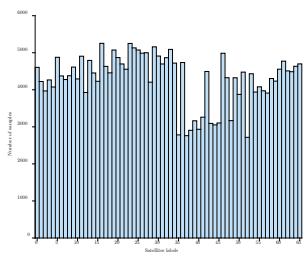

Fig. 5. Sample quantity distribution.

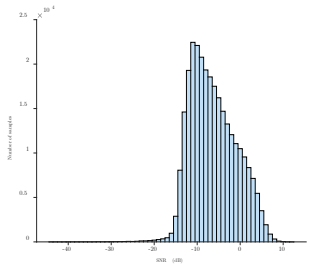

Fig. 6. Signal-to-noise ratio (SNR) distribution.

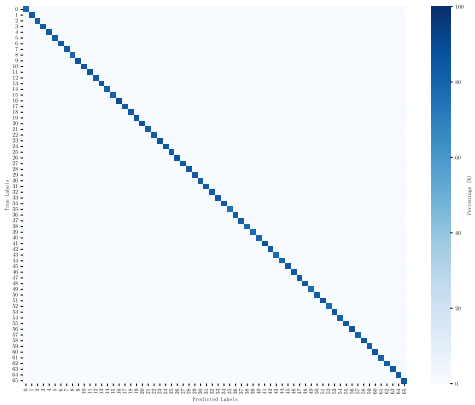

Fig. 7. Confusion matrix of the 66-class using the UW segments.

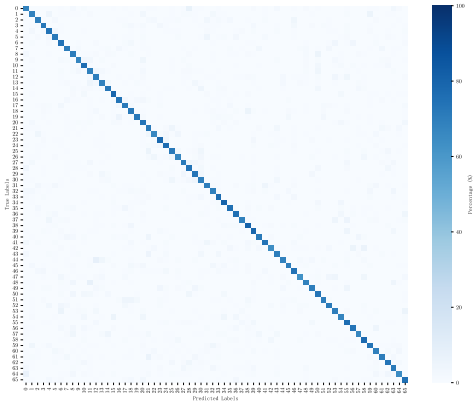

Fig. 8. Confusion matrix of the 66-class using the QPSK segments.

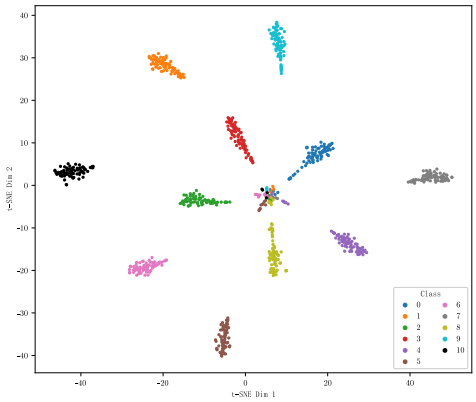

Fig. 9. t-SNE visualization of the feature embeddings for the 66 Iridium satellites based on the UW segments.

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